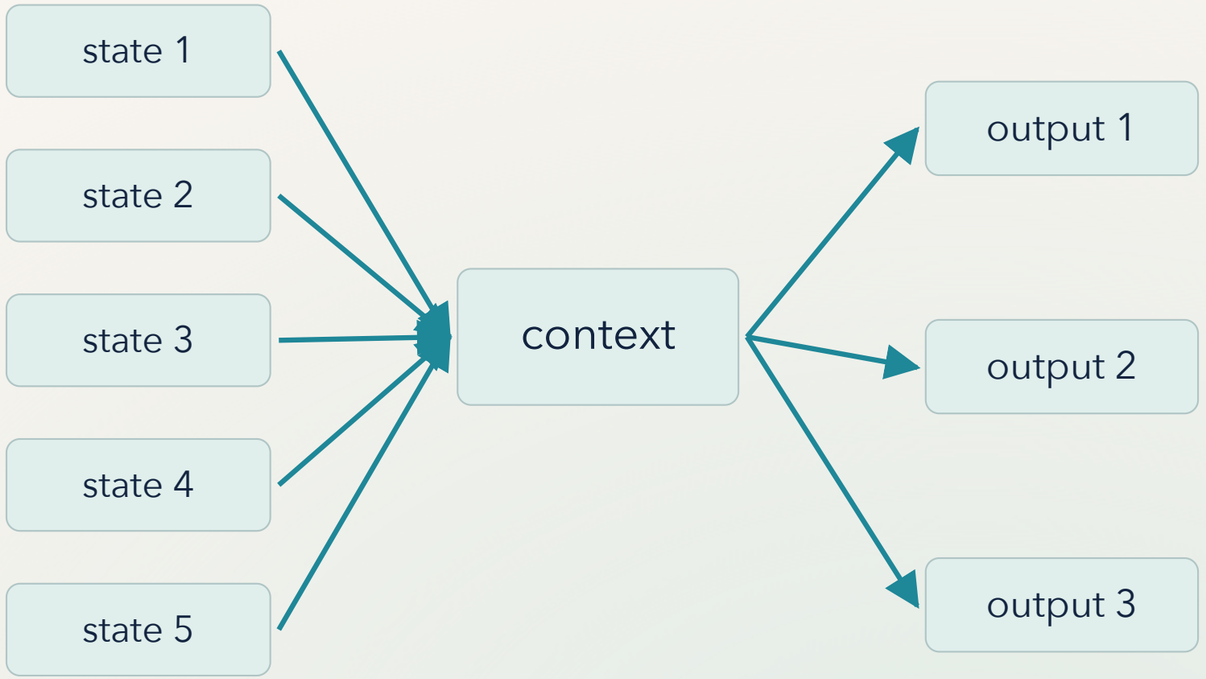
Retrieve what the current output needs

Attention computes a query-dependent combination of available values.

> `query → relevance weights → retrieved context`



Which information reaches each output?

Mechanism diagram / illustrative teaching values

One context can be a bottleneck

Different output positions may need different parts of the input.

A

All outputs share one final context

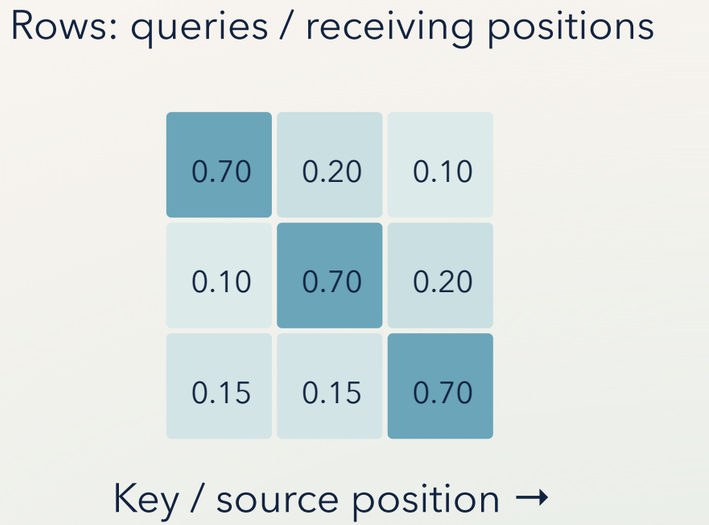
B

Each output queries encoder states

Mechanism diagram / illustrative teaching values

Alignment is output-dependent

A translation decoder may emphasize different source positions at different steps.

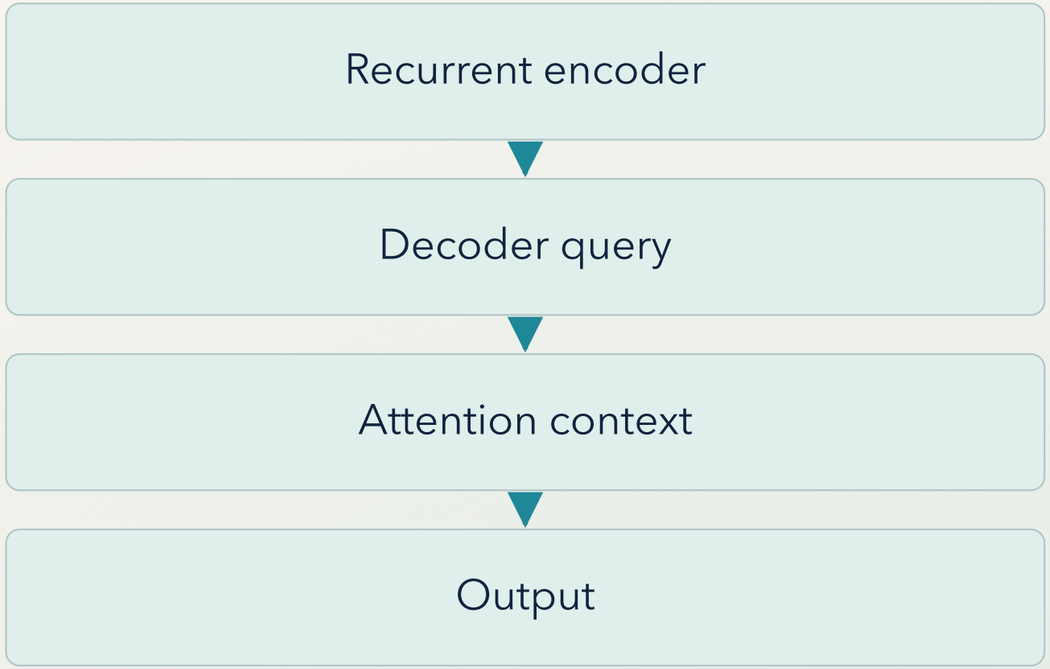


Mechanism diagram / illustrative teaching values

Attention predates the Transformer

LSTM or GRU encoders can supply states to an attention mechanism.

> 2014/15: recurrent attention 2017: Transformer



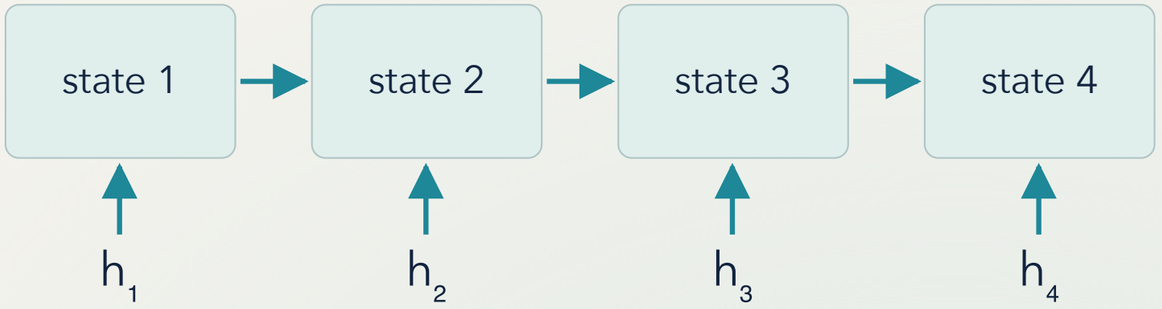
Mechanism diagram / illustrative teaching values

Keep the full set of encoder states

The decoder can retrieve from H instead of using only the final state.

$$H = [h_1; h_2; \ldots; h_n]$$

Shared update parameters across time



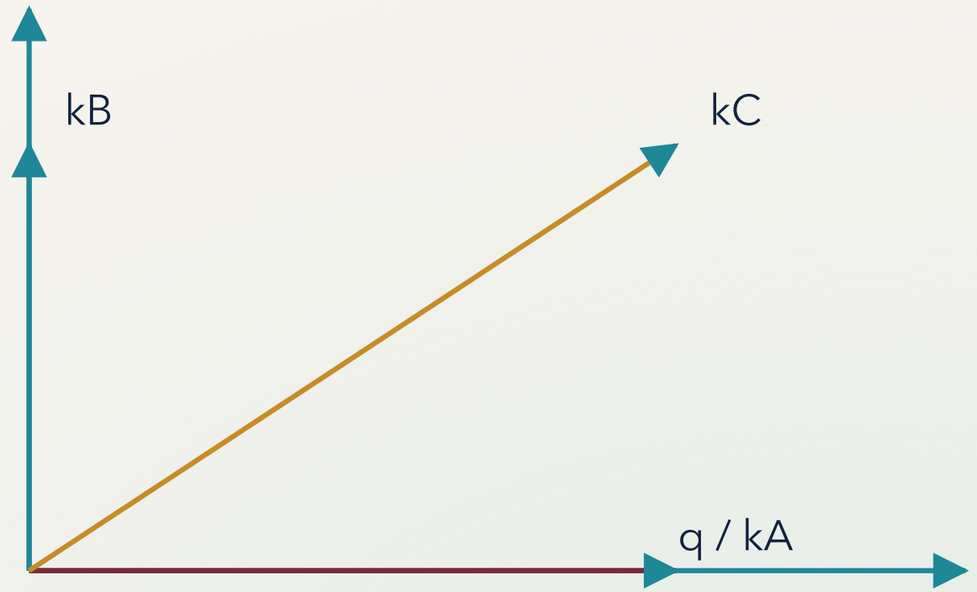
Past → running state → later state

Mechanism diagram / illustrative teaching values

A query specifies what to look for

The query is a vector used to compare the current need with memory keys.

$$q \in \mathbb{R}^{d_k}$$



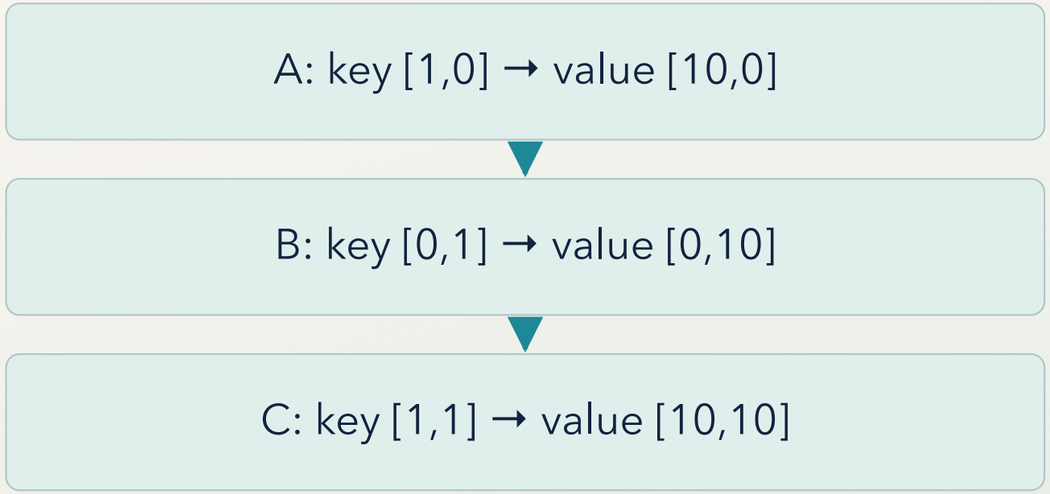
Chosen vectors for an inspectable example

Mechanism diagram / illustrative teaching values

Keys address; values carry content

Matching uses keys; the retrieved combination uses values.

$$K \in \mathbb{R}^{N \times d_k} \quad V \in \mathbb{R}^{N \times d_v}$$

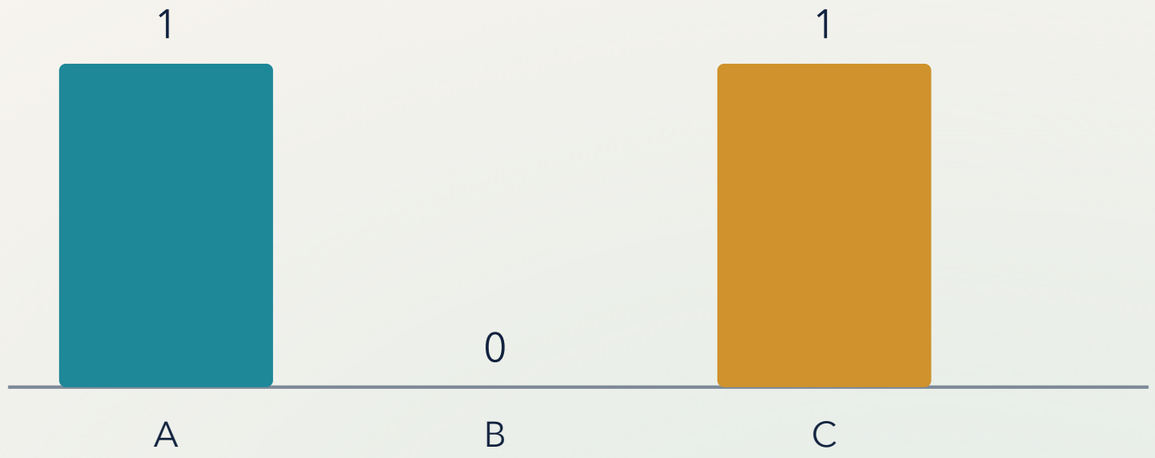


Mechanism diagram / illustrative teaching values

A score is a compatibility function

A score compares one query with one key before normalization.

> dot score: $e_j = q \cdot k_j$



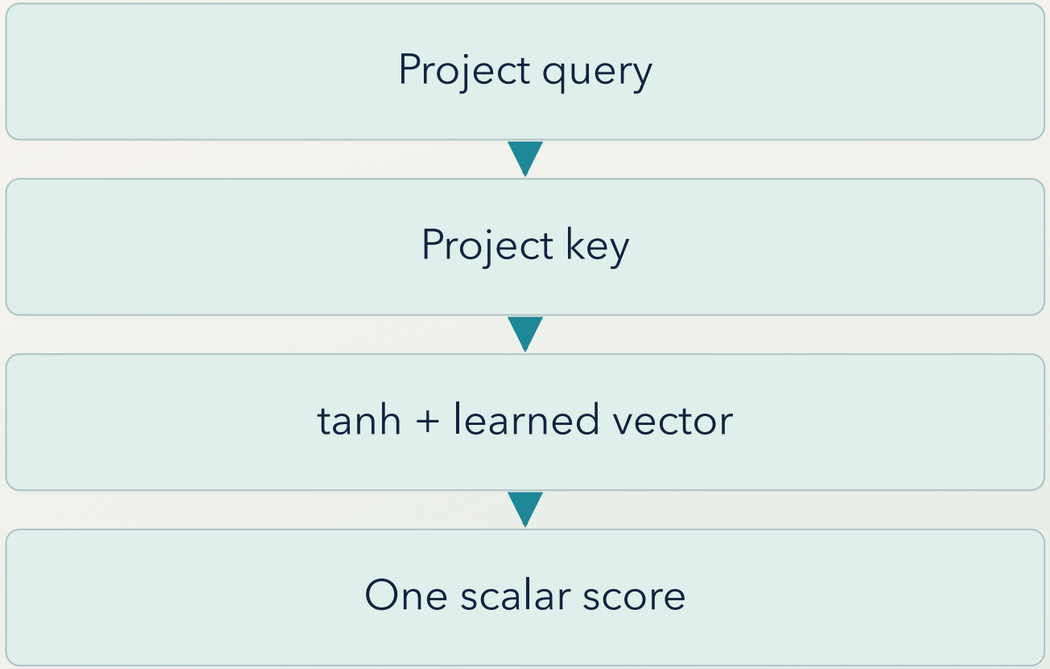
Illustrative values; see the stated calculation

Mechanism diagram / illustrative teaching values

Additive attention learns a scoring network

An MLP can compare a decoder query with each encoder key.

$$e_{ij} = v_a^T \tanh(W_q q_i + W_k k_j + b)$$

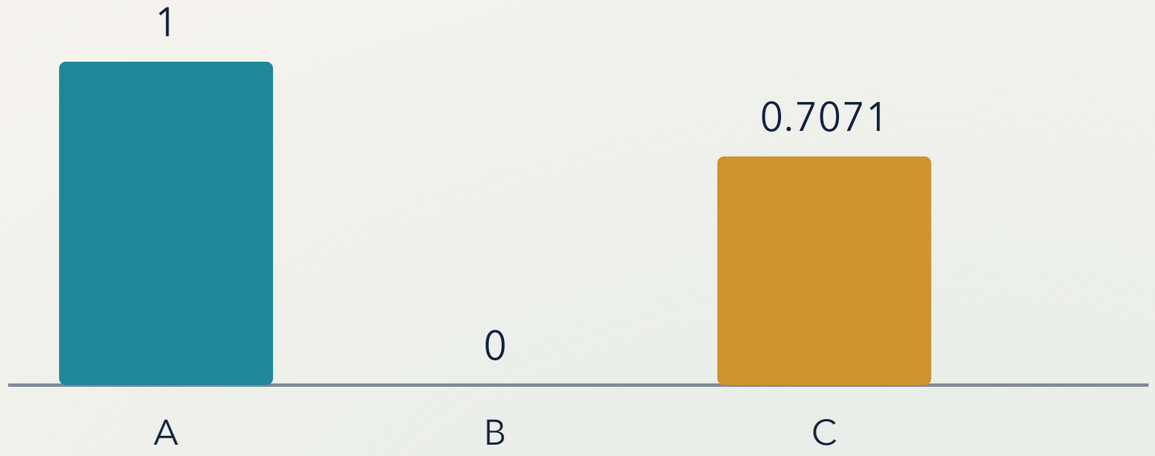


Mechanism diagram / illustrative teaching values

Scaled dot products control score magnitude

Divide by $\sqrt{d_k}$ so high-dimensional dot products are less likely to saturate softmax.

$$S = QK^T / \sqrt{d_k}$$



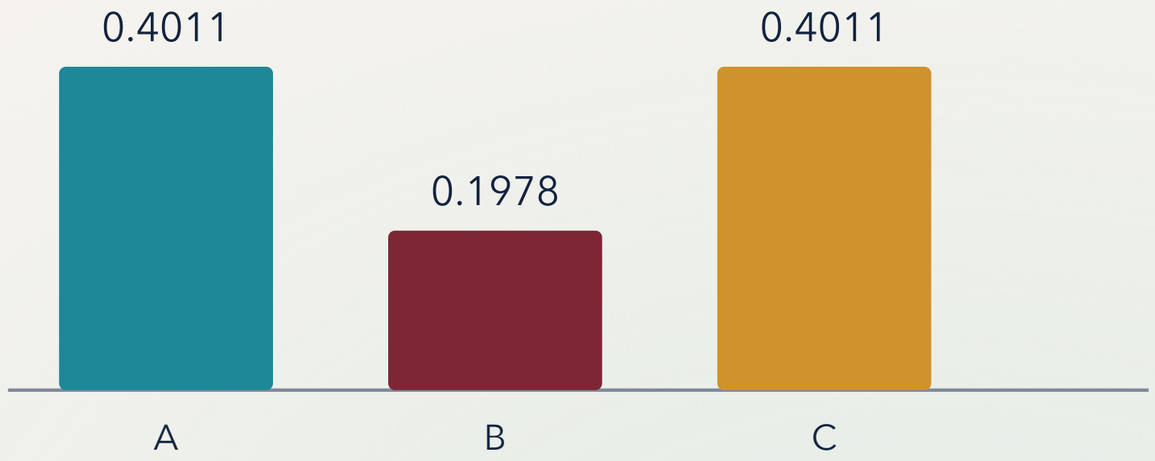
Illustrative values; see the stated calculation

Mechanism diagram / illustrative teaching values

Softmax allocates a unit total weight

Normalize separately for each query over its available keys.

$$a_{ij} = \exp(s_{ij}) \,/\, \Sigma_m \exp(s_{im})$$



Illustrative values; see the stated calculation

Mechanism diagram / illustrative teaching values

Retrieval is a weighted sum

Each value contributes according to its attention weight.

$$c_i = \sum_j a_{ij} v_j$$

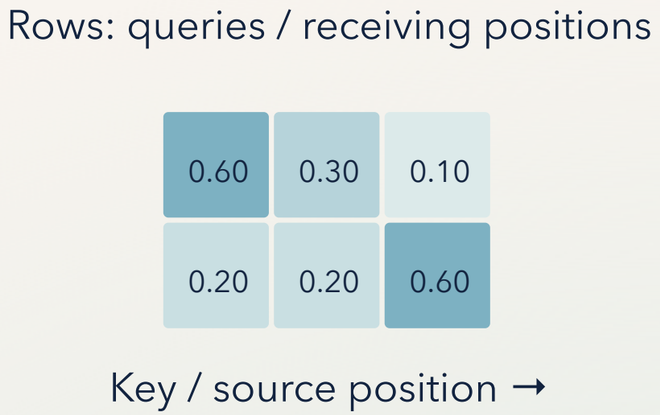


Mechanism diagram / illustrative teaching values

Read matrix axes before the formula

Rows are query destinations; columns are key sources.

$$(Nq \times d_k)(d_k \times Nk) \rightarrow Nq \times Nk$$

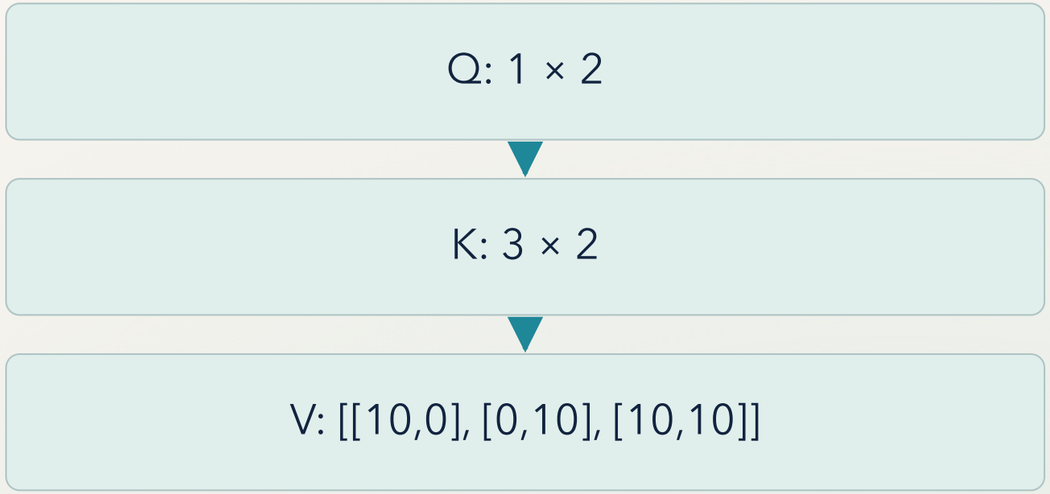


Mechanism diagram / illustrative teaching values

Work one example: choose Q, K, and V

The values are deliberately small so all intermediates can be checked.

> `q=[1,0] K=[[1,0],[0,1],[1,1]]`



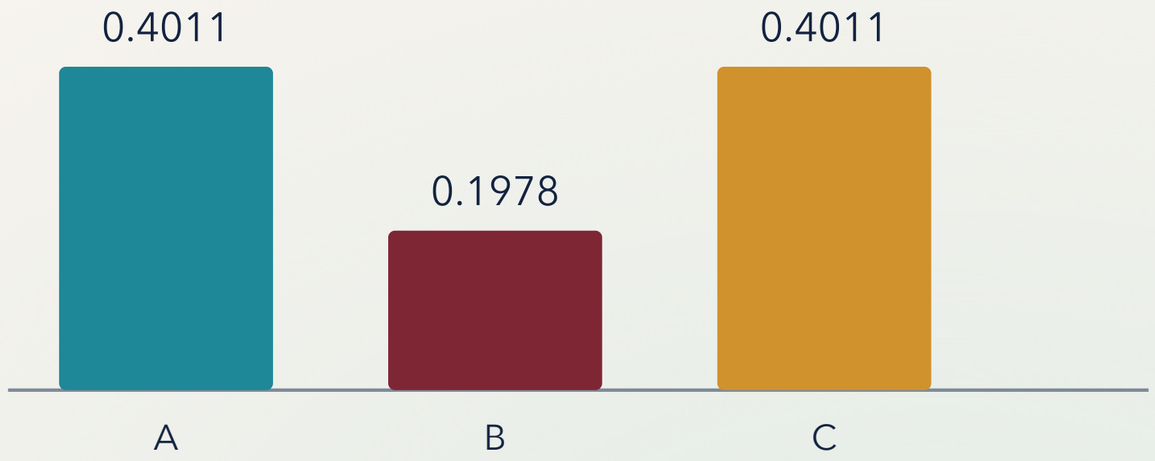
Mechanism diagram / illustrative teaching values

Work one example: scores become weights

A zero dot product still receives a positive normalized weight.

> `scores ≈ [0.7071, 0, 0.7071]`

What are the three attention weights?



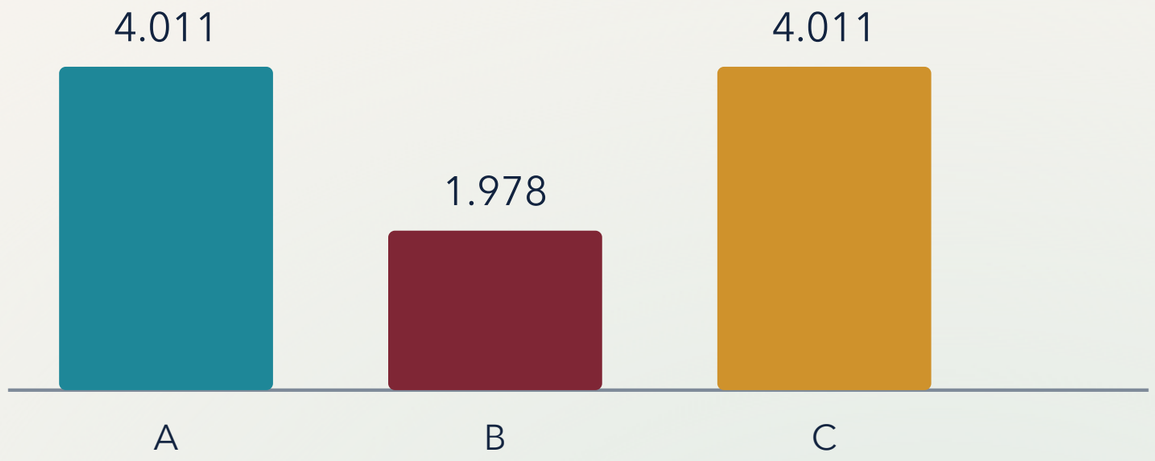
Illustrative values; see the stated calculation

Mechanism diagram / illustrative teaching values

Work one example: combine value coordinates

Use the same weights for every value coordinate.

> `context ≈ [8.0222, 5.9889]`

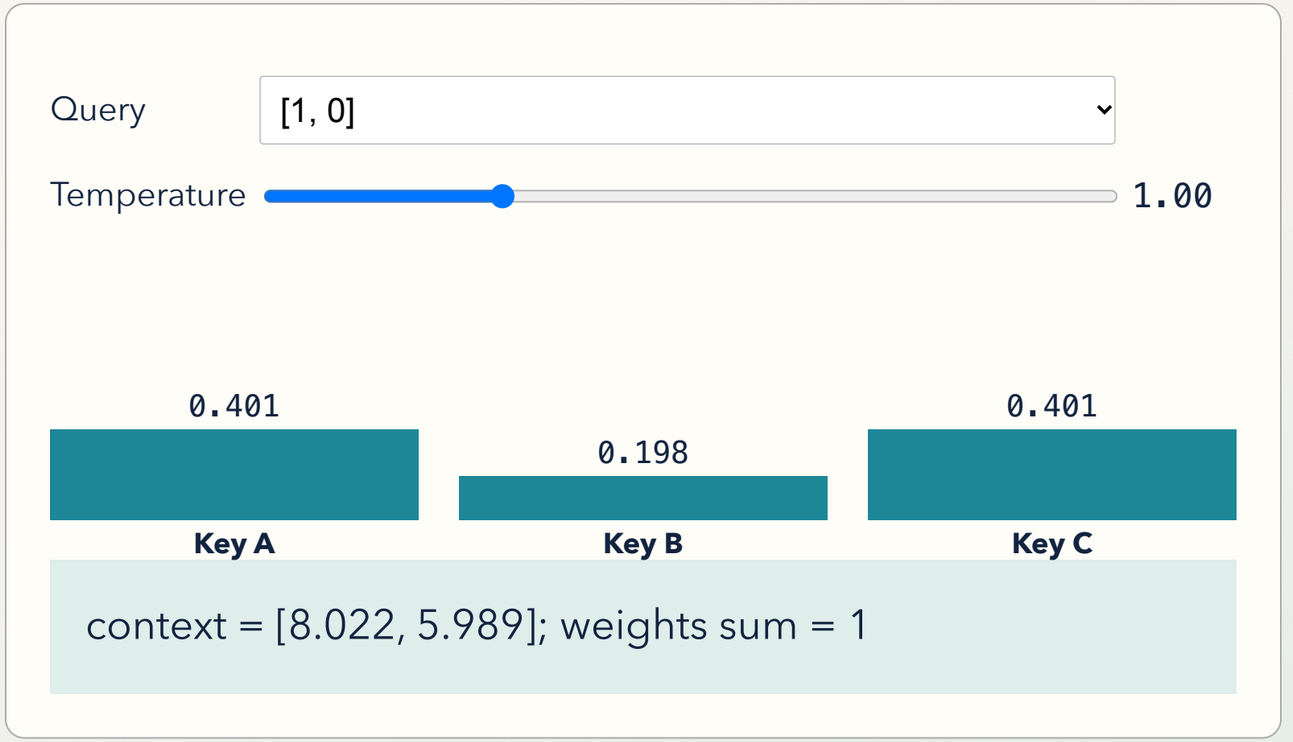


Illustrative values; see the stated calculation

Mechanism diagram / illustrative teaching values

Change the query; keep memory fixed

Attention can route the same memory differently for each request.

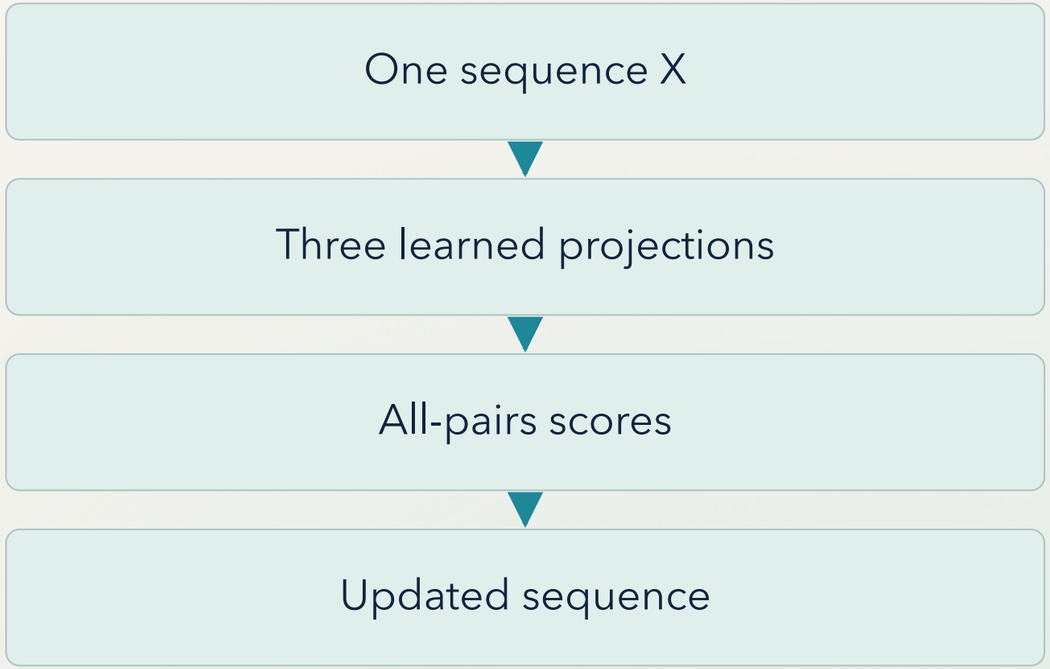


Interactive chosen values — change one variable

Self-attention uses one source sequence

Q, K, and V come from learned projections of the same representations.

> `Q=XWQ K=XWK V=XWV`

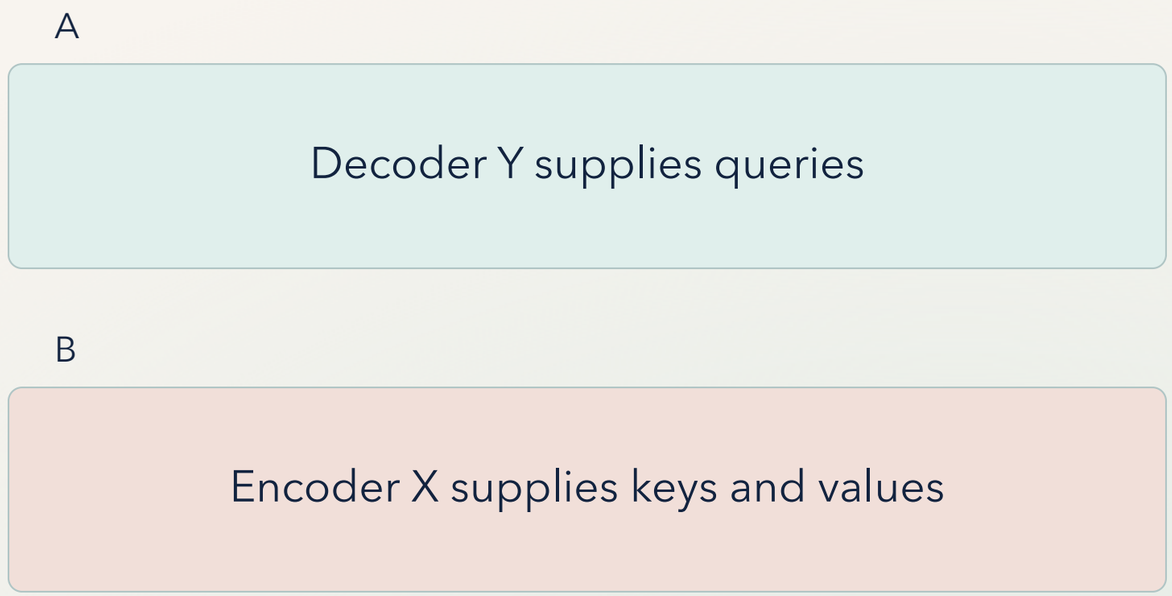


Mechanism diagram / illustrative teaching values

Cross-attention connects two streams

Queries come from one sequence; keys and values come from another.

> `Q=YWQ K=XWK V=XWV`

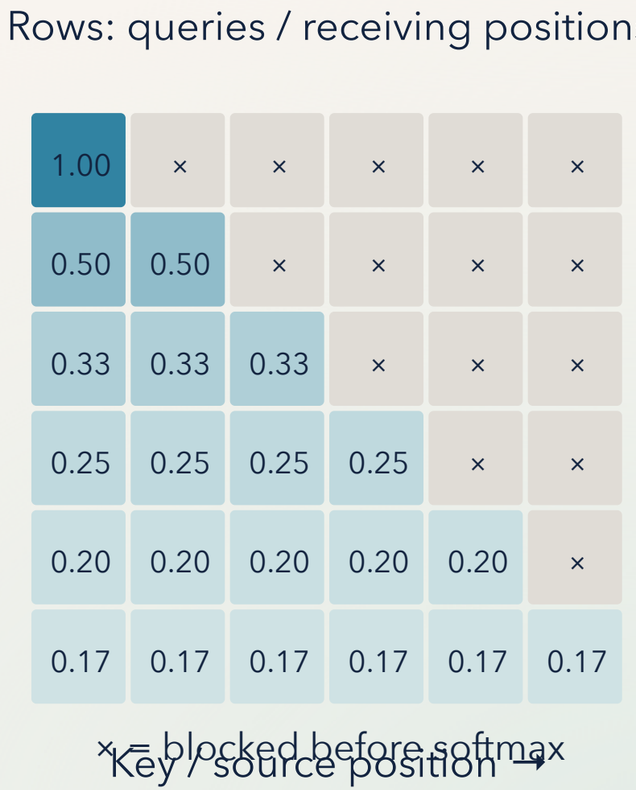


Mechanism diagram / illustrative teaching values

Causal masking hides future keys

Before softmax, set scores for key positions j>i to −∞.

> `allowed(i,j) ⇔ j ≤ i`



Mechanism diagram / illustrative teaching values

Padding masks hide nonexistent tokens

Padding and causality enforce different constraints.

Rows: queries / receiving positions

1.00	×	×	×	×	×
0.50	0.50	×	×	×	×
0.33	0.33	0.33	×	×	×
0.25	0.25	0.25	0.25	×	×
0.20	0.20	0.20	0.20	×	×
0.17	0.17	0.17	0.17	×	×

× = blocked before softmax

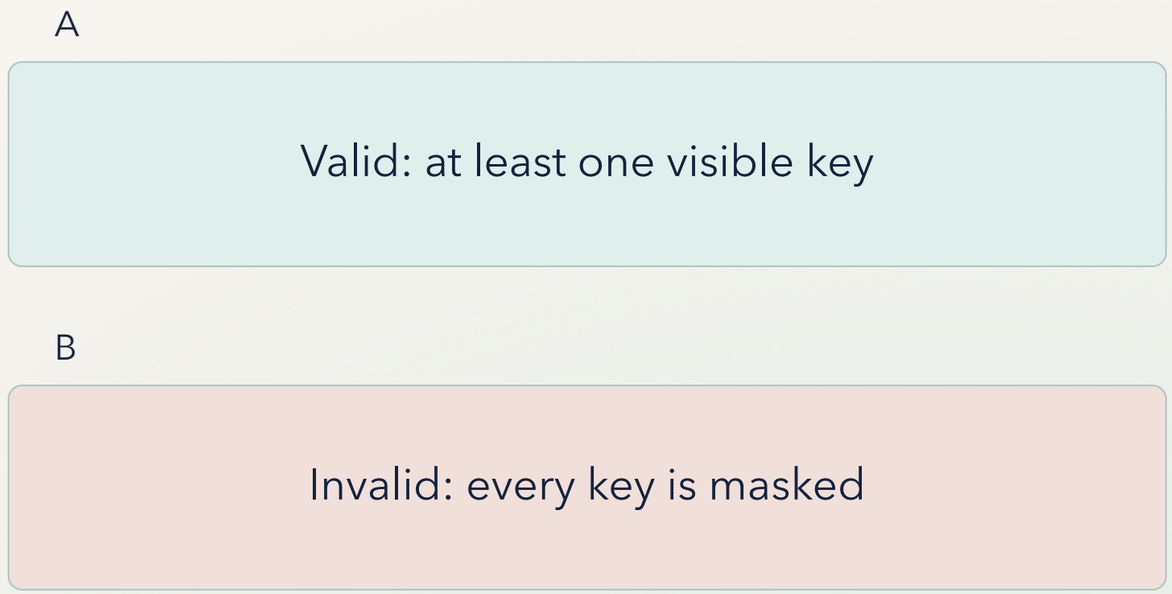
Key / source position →

Mechanism diagram / illustrative teaching values

An all-masked row is undefined

Softmax requires at least one permitted source per query.

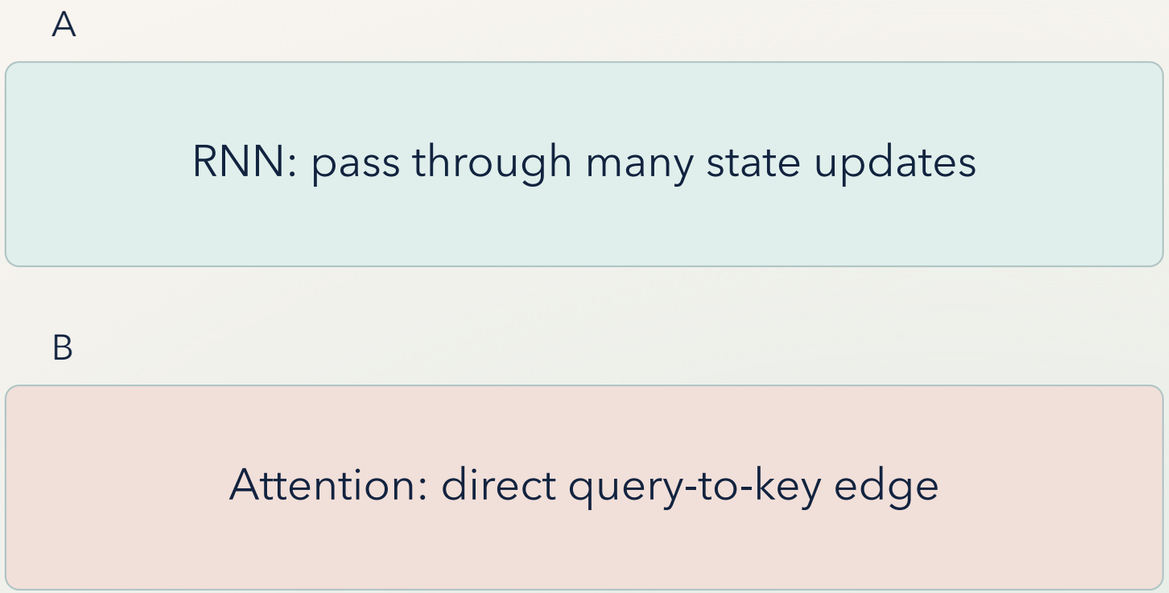
> `all scores = −∞ → no valid normalization`



Mechanism diagram / illustrative teaching values

Attention exposes a short information path

A query can access a distant key within one dense attention layer.

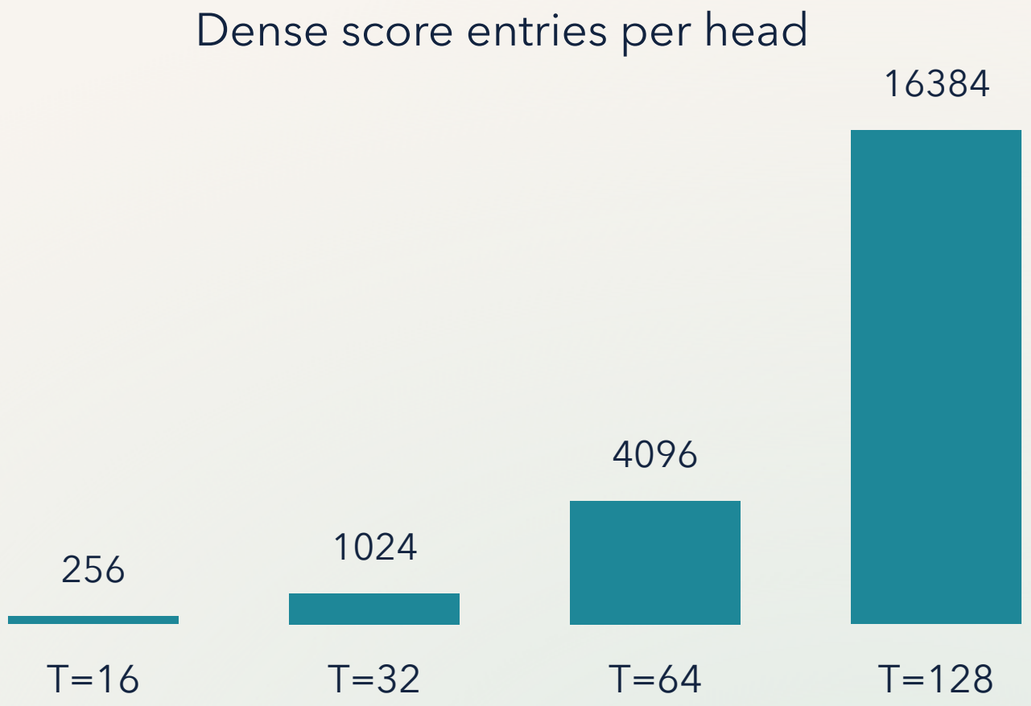


Mechanism diagram / illustrative teaching values

Dense attention has a quadratic score grid

Doubling context length quadruples score entries at fixed head count.

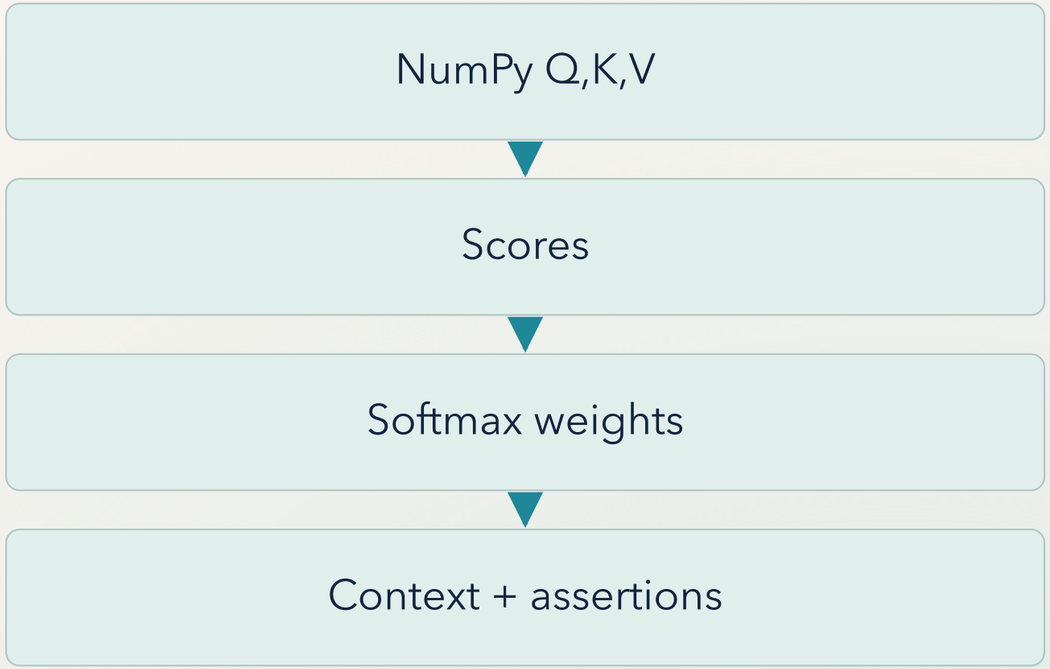
> `per head: T × T scores`



Mechanism diagram / illustrative teaching values

Open the attention notebook

Verify your hand calculation before exploring masks.

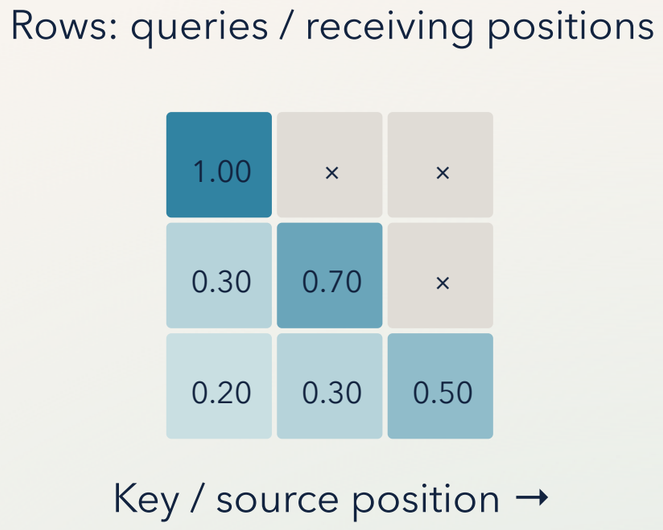


Mechanism diagram / illustrative teaching values

Test causality and normalization

Every permitted row sums to one; every future weight is zero.

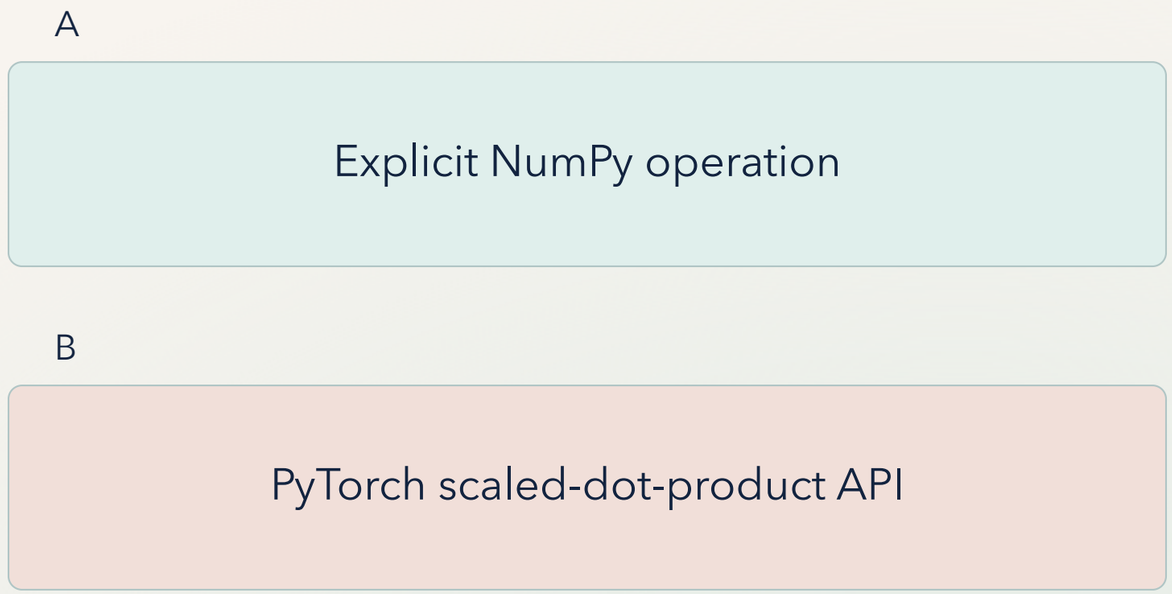
> `row_sum(A)=1 A[i,j>i]=0`



Mechanism diagram / illustrative teaching values

Compare two implementations of one operation

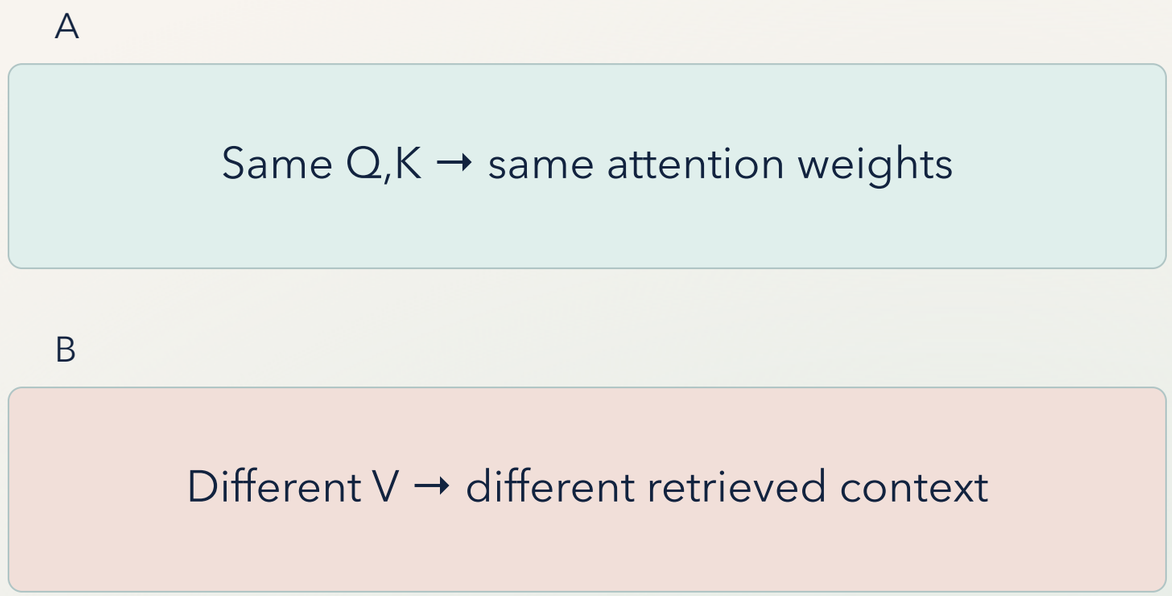
NumPy and PyTorch should agree for the same inputs, scale, and mask.



Mechanism diagram / illustrative teaching values

Routing weights are not causal explanations

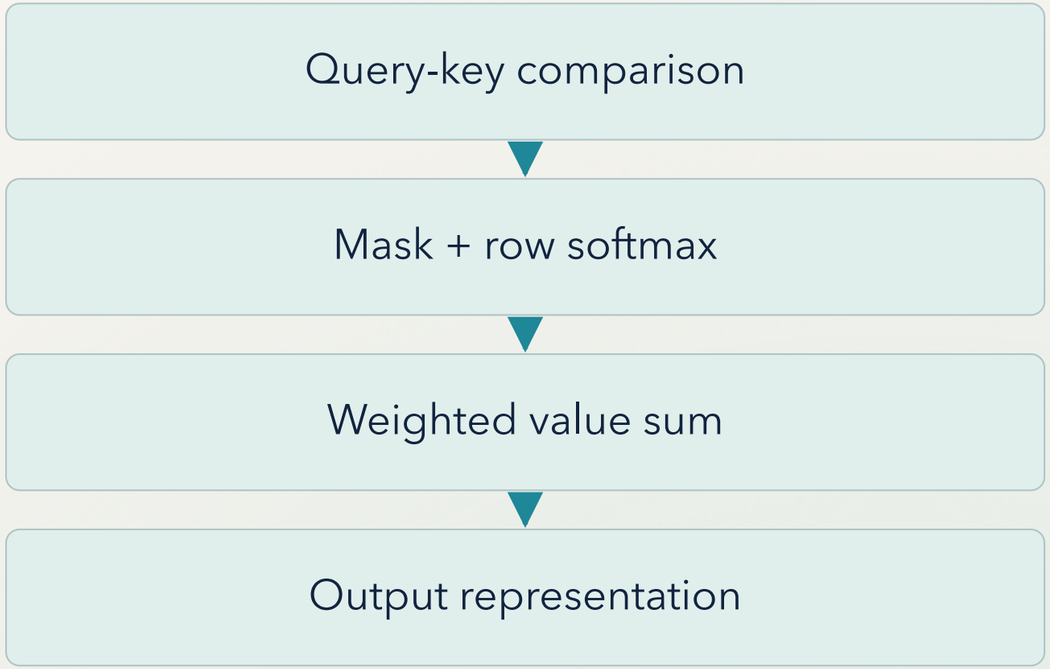
Changing values can alter the output while attention weights remain fixed.



Mechanism diagram / illustrative teaching values

Explain the mechanism before the architecture

State the sequence: compare, normalize, retrieve.

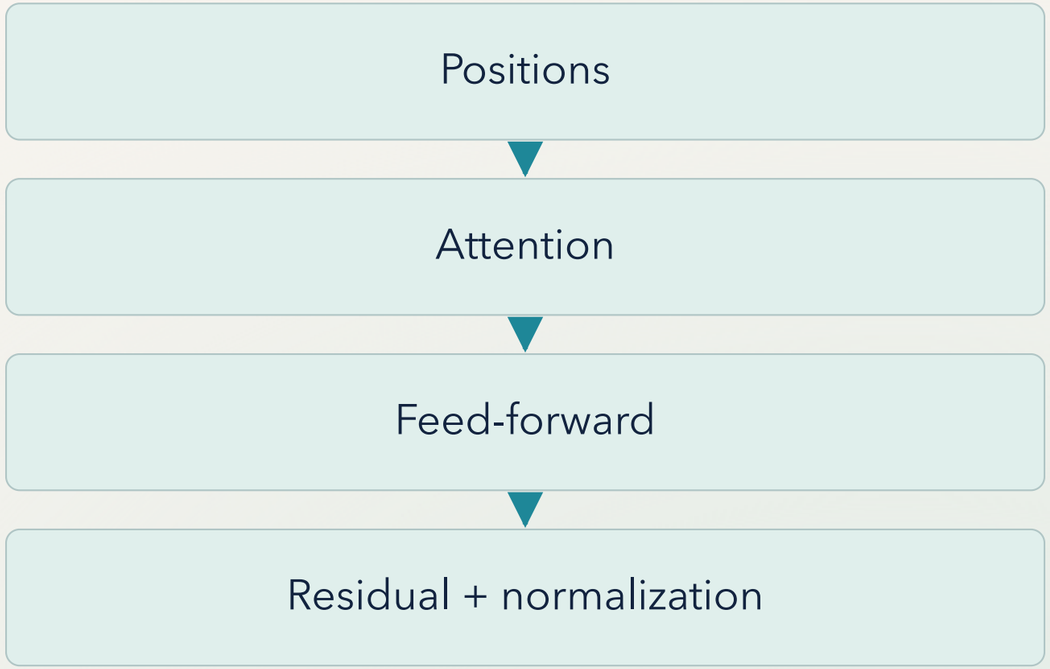


Mechanism diagram / illustrative teaching values

Exit: attention needs an architecture around it

Position information, nonlinear transformations, residual paths, and normalization complete the next story.

Why can an LSTM already use attention?



Mechanism diagram / illustrative teaching values